Day 68 — Final Validation

Day 68 — Final Optimization, Bug Fixing and Validation Report

Scope

Reviewed the latest supplied repository snapshot, checked key pipeline modules, ran the repository tests in the available isolated environment, and added a focused end-to-end regression test for all synthetic demo candidates plus recruiter-report formatting.

Changes in this batch

• Added `tests/test_day68_final_validation.py` to exercise each synthetic candidate through the hiring pipeline simulator and report formatter.

• Added `scripts/run_final_demo.py` as a repeatable, deterministic demonstration entry point.

• No existing score weights, decision thresholds, parser logic or production dependencies were changed.

• No existing defects were silently rewritten without evidence.

Validation results on the supplied repository snapshot

The one warning in the passing subset is the pre-existing `PytestReturnNotNoneWarning` from `tests/test_ats_system_evaluation.py`. The complete suite result is not represented as green because collection was blocked by the missing dependency.

Observed limitations (not fixed in this batch)

• `ats_engine/api.py` uses in-memory resume/job stores and describes parser integration as a boundary; it is not durable production job processing.

• `HiringPipelineSimulator` is a deterministic test workflow using structured stage inputs. Its documented scope excludes actual resume upload/extraction transport.

• `MachineTestEngine` evaluates supplied execution/test results; it does not run candidate-submitted code.

• Recommendation confidence is coverage-based heuristic output, not a calibrated probability.

• Observability and audit utilities are in-process foundations, not a persistent monitoring/audit deployment.

• The scoring modules include rule/rubric-based heuristics; do not represent all modules as trained AI models.

Final status

The supplied repository passes 396 tests in the available environment when the two tests that cannot import `sentence_transformers` are excluded. Run `python -m pytest -v` in the project virtual environment with all requirements installed to establish the definitive local result. The new focused Day 68 test must also pass before release tagging or final submission.